

Creation Date 07-Apr-2025

Revision Date 07-Apr-2025

Revision Number 1

## Section 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: **Benzoyl peroxide / Dicyclohexyl phthalate blend (50/50)**  
Cat No. : **S36107**

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address [begel.sdsdesk@thermofisher.com](mailto:begel.sdsdesk@thermofisher.com)

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## Section 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

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|                                     |                     |
|-------------------------------------|---------------------|
| Organic peroxides                   | Type B (H241)       |
| <b><u>Health hazards</u></b>        |                     |
| Serious Eye Damage/Eye Irritation   | Category 2 (H319)   |
| Skin Sensitization                  | Category 1 (H317)   |
| Reproductive Toxicity               | Category 1B (H360D) |
| <b><u>Environmental hazards</u></b> |                     |
| Chronic aquatic toxicity            | Category 3 (H412)   |

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

### Hazard Statements

- H241 - Heating may cause a fire or explosion
- H317 - May cause an allergic skin reaction
- H319 - Causes serious eye irritation
- H360D - May damage the unborn child
- H412 - Harmful to aquatic life with long lasting effects

### Precautionary Statements

- P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking
- P280 - Wear protective gloves/protective clothing/eye protection/face protection
- P370 + P380 + P375 - In case of fire: Evacuate area. Fight fire remotely due to the risk of explosion
- P302 + P352 - IF ON SKIN: Wash with plenty of soap and water
- P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
- P308 + P313 - IF exposed or concerned: Get medical advice/attention

### Additional EU labelling

Restricted to professional users

## 2.3. Other hazards

- Contains a known or suspected endocrine disruptor
- Included in the list established in accordance with Article 59(1) for having endocrine disrupting properties
- Contains a substance on the National Authorities Endocrine Disruptor Lists

## Section 3: Composition/information on ingredients

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## 3.2. Mixtures

| Component                                        | CAS No  | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                   |
|--------------------------------------------------|---------|-------------------|----------|---------------------------------------------------------------------|
| Dibenzoyl peroxide                               | 94-36-0 | EEC No. 202-327-6 | 50       | Org. Perox. B (H241)<br>Eye Irrit. 2 (H319)<br>Skin Sens. 1 (H317)  |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | 84-61-7 | EEC No. 201-545-9 | 50       | Skin Sens. 1 (H317)<br>Repr. 1B (H360D)<br>Aquatic Chronic 3 (H412) |

Full text of Hazard Statements: see section 16

## Section 4: First aid measures

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | If symptoms persist, call a physician.                                                                                                           |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                            |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable. . May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

### 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                        |
|---------------------------|------------------------|
| <b>Notes to Physician</b> | Treat symptomatically. |
|---------------------------|------------------------|

## Section 5: Firefighting measures

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Water spray, carbon dioxide (CO2), dry chemical, alcohol-resistant foam.

#### Extinguishing media which must not be used for safety reasons

No information available.

### 5.2. Special hazards arising from the substance or mixture

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Thermal decomposition can lead to release of irritating gases and vapors.

## Hazardous Combustion Products

Carbon oxides.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system.

### 6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7: Handling and storage

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 4.1A

**Switzerland - Storage of hazardous substances**

Storage class - SC 4.1  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

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## Section 8: Exposure controls/personal protection

### 8.1. Control parameters

#### Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component                                        | European Union | The United Kingdom                                                 | France                                     | Belgium                         | Spain                                       |
|--------------------------------------------------|----------------|--------------------------------------------------------------------|--------------------------------------------|---------------------------------|---------------------------------------------|
| Dibenzoyl peroxide                               |                | STEL: 15 mg/m <sup>3</sup> 15 min<br>TWA: 5 mg/m <sup>3</sup> 8 hr | TWA / VME: 5 mg/m <sup>3</sup> (8 heures). | TWA: 5 mg/m <sup>3</sup> 8 uren | TWA / VLA-ED: 5 mg/m <sup>3</sup> (8 horas) |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester |                | STEL: 15 mg/m <sup>3</sup> 15 min<br>TWA: 5 mg/m <sup>3</sup> 8 hr |                                            |                                 |                                             |

| Component          | Italy | Germany                                                                                                                                                                                                                                                                                                                     | Portugal                         | The Netherlands | Finland                                                                          |
|--------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------|----------------------------------------------------------------------------------|
| Dibenzoyl peroxide |       | TWA: 5 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 1<br>TWA: 4 mg/m <sup>3</sup> (8 Stunden). MAK can occur as vapor and aerosol at the same time<br>TWA: 1 mg/m <sup>3</sup> (8 Stunden). MAK can occur as vapor and aerosol at the same time<br>Höhepunkt: 8 mg/m <sup>3</sup><br>Höhepunkt: 4 mg/m <sup>3</sup> | TWA: 5 mg/m <sup>3</sup> 8 horas |                 | TWA: 5 mg/m <sup>3</sup> 8 tunteina<br>STEL: 10 mg/m <sup>3</sup> 15 minuutteina |

| Component                                        | Austria                                                                             | Denmark                                                                    | Switzerland                                                                | Poland                                                                         | Norway                                                                                       |
|--------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Dibenzoyl peroxide                               | MAK-KZGW: 10 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 5 mg/m <sup>3</sup> 8 Stunden | TWA: 5 mg/m <sup>3</sup> 8 timer<br>STEL: 10 mg/m <sup>3</sup> 15 minutter | STEL: 5 mg/m <sup>3</sup> 15 Minuten<br>TWA: 5 mg/m <sup>3</sup> 8 Stunden | STEL: 10 mg/m <sup>3</sup> 15 minutach<br>TWA: 5 mg/m <sup>3</sup> 8 godzinach | TWA: 5 mg/m <sup>3</sup> 8 timer<br>STEL: 10 mg/m <sup>3</sup> 15 minutter. value calculated |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | MAK-TMW: 5 mg/m <sup>3</sup> 8 Stunden                                              | TWA: 3 mg/m <sup>3</sup> 8 timer<br>STEL: 6 mg/m <sup>3</sup> 15 minutter  |                                                                            |                                                                                |                                                                                              |

| Component                                        | Bulgaria | Croatia                                | Ireland                                                             | Cyprus | Czech Republic                                                        |
|--------------------------------------------------|----------|----------------------------------------|---------------------------------------------------------------------|--------|-----------------------------------------------------------------------|
| Dibenzoyl peroxide                               |          | TWA-GVI: 5 mg/m <sup>3</sup> 8 satima. | TWA: 5 mg/m <sup>3</sup> 8 hr.<br>STEL: 15 mg/m <sup>3</sup> 15 min |        | TWA: 5 mg/m <sup>3</sup> 8 hodinách.<br>Ceiling: 10 mg/m <sup>3</sup> |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester |          | TWA-GVI: 5 mg/m <sup>3</sup> 8 satima. | TWA: 5 mg/m <sup>3</sup> 8 hr.<br>STEL: 15 mg/m <sup>3</sup> 15 min |        |                                                                       |

| Component                                        | Estonia                              | Gibraltar | Greece                   | Hungary                                                                                                                     | Iceland                                                                                                                                                                |
|--------------------------------------------------|--------------------------------------|-----------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dibenzoyl peroxide                               | TWA: 5 mg/m <sup>3</sup> 8 tundides. |           | TWA: 5 mg/m <sup>3</sup> | STEL: 5 mg/m <sup>3</sup> 15 percekben. CK<br>TWA: 5 mg/m <sup>3</sup> 8 órában. AK lehetséges borön keresztüli felszívódás | TWA: 5 mg/m <sup>3</sup> 8 klukkustundum.<br>Ceiling: 10 mg/m <sup>3</sup>                                                                                             |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester |                                      |           |                          |                                                                                                                             | TWA: 3 mg/m <sup>3</sup> 8 klukkustundum. same limit value expressed in mg/m <sup>3</sup> shall also be applied for those phthalates for which no limit value has been |

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|  |  |  |  |  |                                                                                                                                                                                                |
|--|--|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  |  |  | defined<br>Ceiling: 6 mg/m <sup>3</sup> same<br>limit value expressed in<br>mg/m <sup>3</sup> shall also be<br>applied for those<br>phthalates for which no<br>limit value has been<br>defined |
|--|--|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Component          | Russia | Slovak Republic          | Slovenia                                                                                                               | Sweden | Turkey |
|--------------------|--------|--------------------------|------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Dibenzoyl peroxide |        | TWA: 5 mg/m <sup>3</sup> | TWA: 5 mg/m <sup>3</sup> 8 urah<br>inhalable fraction<br>STEL: 5 mg/m <sup>3</sup> 15<br>minutah inhalable<br>fraction |        |        |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                                                             | Acute effects local<br>(Dermal) | Acute effects<br>systemic (Dermal) | Chronic effects local<br>(Dermal) | Chronic effects<br>systemic (Dermal) |
|-----------------------------------------------------------------------|---------------------------------|------------------------------------|-----------------------------------|--------------------------------------|
| Dibenzoyl peroxide<br>94-36-0 ( 50 )                                  |                                 |                                    | DNEL = 34µg/cm <sup>2</sup>       | DNEL = 13.3mg/kg<br>bw/day           |
| 1,2-Benzenedicarboxylic acid,<br>dicyclohexyl ester<br>84-61-7 ( 50 ) |                                 |                                    |                                   | DNEL = 0.5mg/kg<br>bw/day            |

| Component                                                             | Acute effects local<br>(Inhalation) | Acute effects<br>systemic (Inhalation) | Chronic effects local<br>(Inhalation) | Chronic effects<br>systemic (Inhalation) |
|-----------------------------------------------------------------------|-------------------------------------|----------------------------------------|---------------------------------------|------------------------------------------|
| Dibenzoyl peroxide<br>94-36-0 ( 50 )                                  |                                     |                                        |                                       | DNEL = 39mg/m <sup>3</sup>               |
| 1,2-Benzenedicarboxylic acid,<br>dicyclohexyl ester<br>84-61-7 ( 50 ) |                                     | DNEL = 35.2mg/m <sup>3</sup>           |                                       | DNEL = 35.2mg/m <sup>3</sup>             |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                                                             | Fresh water     | Fresh water<br>sediment              | Water Intermittent | Microorganisms in<br>sewage treatment | Soil (Agriculture)               |
|-----------------------------------------------------------------------|-----------------|--------------------------------------|--------------------|---------------------------------------|----------------------------------|
| Dibenzoyl peroxide<br>94-36-0 ( 50 )                                  | PNEC = 0.02µg/L | PNEC =<br>0.0127mg/kg<br>sediment dw | PNEC = 0.602µg/L   | PNEC = 0.35mg/L                       | PNEC =<br>0.0025mg/kg soil<br>dw |
| 1,2-Benzenedicarboxylic<br>acid, dicyclohexyl ester<br>84-61-7 ( 50 ) | PNEC = 1.04µg/L | PNEC = 1.06mg/kg<br>sediment dw      | PNEC = 0.02mg/L    | PNEC = 10mg/L                         | PNEC = 0.31mg/kg<br>soil dw      |

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| Component                                                          | Marine water     | Marine water sediment              | Marine water Intermittent | Food chain             | Air |
|--------------------------------------------------------------------|------------------|------------------------------------|---------------------------|------------------------|-----|
| Dibenzoyl peroxide<br>94-36-0 ( 50 )                               | PNEC = 0.002µg/L | PNEC = 0.00127mg/kg<br>sediment dw |                           |                        |     |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester<br>84-61-7 ( 50 ) | PNEC = 0.104µg/L | PNEC = 0.11mg/kg<br>sediment dw    | PNEC = 0.02mg/L           | PNEC = 133g/kg<br>food |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material                                      | Breakthrough time                    | Glove thickness | EU standard | Glove comments        |
|-----------------------------------------------------|--------------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber<br>Neoprene<br>Natural rubber<br>PVC | See manufacturers<br>recommendations | -               | EN 374      | (minimum requirement) |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

#### Physical State

Solid

#### Appearance

ALFAAS36107

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|                                                  |                          |                                   |
|--------------------------------------------------|--------------------------|-----------------------------------|
| Odor                                             | No information available |                                   |
| Odor Threshold                                   | No data available        |                                   |
| Melting Point/Range                              | 54 °C / 129.2 °F         |                                   |
| Softening Point                                  | No data available        |                                   |
| Boiling Point/Range                              | No information available |                                   |
| Flammability (liquid)                            | Not applicable           | Solid                             |
| Flammability (solid,gas)                         | No information available |                                   |
| Explosion Limits                                 | No data available        |                                   |
| Flash Point                                      | No information available | Method - No information available |
| Autoignition Temperature                         | No data available        |                                   |
| Decomposition Temperature                        | No data available        |                                   |
| pH                                               | Not applicable           |                                   |
| Viscosity                                        | Not applicable           | Solid                             |
| Water Solubility                                 | No information available |                                   |
| Solubility in other solvents                     | No information available |                                   |
| Partition Coefficient (n-octanol/water)          |                          |                                   |
| Component                                        | log Pow                  |                                   |
| Dibenzoyl peroxide                               | 3.2                      |                                   |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | 5.6                      |                                   |
| Vapor Pressure                                   | No data available        |                                   |
| Density / Specific Gravity                       | No data available        |                                   |
| Bulk Density                                     | No data available        |                                   |
| Vapor Density                                    | Not applicable           | Solid                             |
| Particle characteristics                         | No data available        |                                   |

## 9.2. Other information

Evaporation Rate Not applicable - Solid

## Section 10: Stability and reactivity

### 10.1. Reactivity

Yes

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

Hazardous Polymerization No information available.  
Hazardous Reactions None under normal processing.

### 10.4. Conditions to avoid

Incompatible products. Excess heat.

### 10.5. Incompatible materials

None known.

### 10.6. Hazardous decomposition products

Carbon oxides.

## Section 11: Toxicological information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008



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## Product Information

### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

## Toxicology data for the components

| Component                                        | LD50 Oral                             | LD50 Dermal                           | LC50 Inhalation |
|--------------------------------------------------|---------------------------------------|---------------------------------------|-----------------|
| Dibenzoyl peroxide                               | LD50 = 7710 mg/kg ( Rat )             | -                                     | -               |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | LD50 > 2000 mg/kg ( Rat )<br>OECD 423 | LD50 > 2000 mg/kg ( Rat )<br>OECD 402 | -               |

### (b) skin corrosion/irritation;

No data available

### (c) serious eye damage/irritation;

Category 2

### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

Category 1

May cause sensitization by skin contact

### (e) germ cell mutagenicity;

No data available

### (f) carcinogenicity;

No data available

There are no known carcinogenic chemicals in this product

### (g) reproductive toxicity;

Category 1B

### (h) STOT-single exposure;

No data available

### (i) STOT-repeated exposure;

No data available

Target Organs

None known.

### (j) aspiration hazard;

Not applicable

Solid

## Symptoms / effects, both acute and delayed

Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

## 11.2. Information on other hazards

### Endocrine Disrupting Properties

#### Assess endocrine disrupting properties for human health

Substance identified as having endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 Contains a substance on the National Authorities Endocrine Disruptor Lists

| Component | EU National Authorities Endocrine Disruptor Lists - |
|-----------|-----------------------------------------------------|
|-----------|-----------------------------------------------------|

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|                                                                    | Health         |
|--------------------------------------------------------------------|----------------|
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester<br>84-61-7 ( 50 ) | List I List II |

## Section 12: Ecological information

### 12.1. Toxicity

#### Ecotoxicity effects

The product contains following substances which are hazardous for the environment.  
Contains a substance which is: Very toxic to aquatic organisms.

| Component                                        | Freshwater Fish                                               | Water Flea                                  | Freshwater Algae                                              |
|--------------------------------------------------|---------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------|
| Dibenzoyl peroxide                               | LC50: = 0.0602 mg/L, 96h<br>semi-static (Oncorhynchus mykiss) |                                             |                                                               |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | LC50 > 2 mg/L Oryzias latipes, 96h (OECD 203)                 | EC50 > 2 mg/L Daphnia magna, 48h (OECD 202) | EC50 > 2 mg/L Psuedokirchneriella subcapitata, 72h (OECD 201) |

| Component                                        | Microtox                                        | M-Factor |
|--------------------------------------------------|-------------------------------------------------|----------|
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | NOEC > 100 mg/L activated sludge, 3h (OECD 209) |          |

### 12.2. Persistence and degradability

#### Degradation in sewage treatment plant

No information available  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### 12.3. Bioaccumulative potential

No information available

| Component                                        | log Pow | Bioconcentration factor (BCF) |
|--------------------------------------------------|---------|-------------------------------|
| Dibenzoyl peroxide                               | 3.2     | No data available             |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | 5.6     | No data available             |

### 12.4. Mobility in soil

No information available

### 12.5. Results of PBT and vPvB assessment

No data available for assessment.

### 12.6. Endocrine disrupting properties

#### Endocrine Disruptor Information

| Component                                        | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances |
|--------------------------------------------------|------------------------------------------|--------------------------------------------------|
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | Group III Chemical                       |                                                  |

### 12.7. Other adverse effects

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## Section 13: Disposal considerations

### 13.1. Waste treatment methods

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|                                            |                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                                                                                     |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                   |
| <b>European Waste Catalogue (EWC)</b>      | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                                   |
| <b>Other Information</b>                   | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains.                          |
| <b>Switzerland - Waste Ordinance</b>       | Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600<br><a href="https://www.fedlex.admin.ch/eli/cc/2015/891/en">https://www.fedlex.admin.ch/eli/cc/2015/891/en</a> |

## Section 14: Transport information

### IMDG/IMO

|                                         |                                |
|-----------------------------------------|--------------------------------|
| <b>14.1. UN number</b>                  | UN3106                         |
| <b>14.2. UN proper shipping name</b>    | ORGANIC PEROXIDE TYPE D, SOLID |
| <b>Technical Shipping Name</b>          | Benzoyl peroxide               |
| <b>14.3. Transport hazard class(es)</b> | 5.2                            |
| <b>14.4. Packing group</b>              |                                |

### ADR

|                                         |                                |
|-----------------------------------------|--------------------------------|
| <b>14.1. UN number</b>                  | UN3106                         |
| <b>14.2. UN proper shipping name</b>    | ORGANIC PEROXIDE TYPE D, SOLID |
| <b>Technical Shipping Name</b>          | Benzoyl peroxide               |
| <b>14.3. Transport hazard class(es)</b> | 5.2                            |
| <b>14.4. Packing group</b>              |                                |

### IATA

|                                         |                                |
|-----------------------------------------|--------------------------------|
| <b>14.1. UN number</b>                  | UN3106                         |
| <b>14.2. UN proper shipping name</b>    | ORGANIC PEROXIDE TYPE D, SOLID |
| <b>Technical Shipping Name</b>          | Benzoyl peroxide               |
| <b>14.3. Transport hazard class(es)</b> | 5.2                            |
| <b>14.4. Packing group</b>              |                                |

|                                                                      |                                  |
|----------------------------------------------------------------------|----------------------------------|
| <b>14.5. Environmental hazards</b>                                   | No hazards identified            |
| <b>14.6. Special precautions for user</b>                            | No special precautions required. |
| <b>14.7. Maritime transport in bulk according to IMO instruments</b> | Not applicable, packaged goods   |

## Section 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

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## International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                                        | CAS No  | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|--------------------------------------------------|---------|-----------|--------|-----|-------|------|----------|------|------|
| Dibenzoyl peroxide                               | 94-36-0 | 202-327-6 | -      | -   | X     | X    | KE-09889 | X    | X    |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | 84-61-7 | 201-545-9 | -      | -   | X     | X    | KE-02215 | X    | X    |

| Component                                        | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDL | AICS | NZIoC | PICCS |
|--------------------------------------------------|---------|------|-----------------------------------------------|-----|-----|------|-------|-------|
| Dibenzoyl peroxide                               | 94-36-0 | X    | ACTIVE                                        | X   | -   | X    | X     | X     |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | 84-61-7 | X    | ACTIVE                                        | X   | -   | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Authorisation/Restrictions according to EU REACH

| Component                                        | CAS No  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                              | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)                                  |
|--------------------------------------------------|---------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Dibenzoyl peroxide                               | 94-36-0 | -                                                                   | Use restricted. See entry 75.<br>(see link for restriction details)                                                                        | -                                                                                                                                      |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | 84-61-7 | -                                                                   | Use restricted. See entry 30.<br>(see link for restriction details)<br>Use restricted. See entry 75.<br>(see link for restriction details) | SVHC Candidate list - 201-545-9 - Toxic for reproduction (Article 57c)<br>Endocrine disrupting properties (Article 57f - human health) |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

## REACH links

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

## Seveso III Directive (2012/18/EC)

| Component                                        | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|--------------------------------------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Dibenzoyl peroxide                               | 94-36-0 | Not applicable                                                                            | Not applicable                                                                           |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | 84-61-7 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

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Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 94/33/EC on the protection of young people at work

Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

**WGK Classification** Water endangering class = 2 (self classification)

| Component                                        | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                              |
|--------------------------------------------------|---------------------------------------|------------------------------------------------------|
| Dibenzoyl peroxide                               | WGK2                                  | Class I : 20 mg/m <sup>3</sup> (Massenkonzentration) |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | WGK1                                  |                                                      |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                                                          | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Dibenzoyl peroxide<br>94-36-0 ( 50 )                               | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester<br>84-61-7 ( 50 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## Section 16: Other information

### Full text of H-Statements referred to under sections 2 and 3

H241 - Heating may cause a fire or explosion

H317 - May cause an allergic skin reaction

H319 - Causes serious eye irritation

H360D - May damage the unborn child

H412 - Harmful to aquatic life with long lasting effects

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

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**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Prepared By** Health, Safety and Environmental Department

**Creation Date** 07-Apr-2025

**Revision Date** 07-Apr-2025

**Revision Summary** Initial Release.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**